SkillBridge: Student Project Discovery & Complementary Skill Matchmaker

One Unified Campus Platform Connecting Project Discovery, Student Skills & Team Sprint Tracking

PROJECT CATEGORY

Academic Minor Project Defense

Department of Computer Science & Engineering

KEY CORE MODULES

Discover • Match • Recruit • Kanban

Ready for Major Project Scaling

Problem Statement & Campus Challenges

Students often have groundbreaking project ideas but fail to execute due to three systemic bottlenecks:



1. Redundant Skill Teams

Student groups form based on friend circles rather than complementary competencies, resulting in teams with 4 backend coders and zero UI/UX or data specialists.



2. Fragmented Discovery

Project recruitment happens across scattered WhatsApp, Telegram, or Discord chats with no structured tracking of domain categories or role requirements.



3. The Execution Abyss

After team formation, students lack dedicated sprint tracking tools. Milestones are neglected, resulting in rushed submissions right before final exam evaluations.

Proposed Solution: The 3 Core Pillars

A unified academic pipeline linking idea discovery, complementary skill recruitment, and sprint execution:



Pillar 1: Idea Pitching & Discovery

Leaders post projects with defined domain tags, submission deadlines, and precise unfilled roles (e.g. "Need: React Dev + Figma Designer").



Pillar 2: Complementary Matching

Synergy algorithm evaluates student competencies against unfilled roles, displaying a dynamic compatibility percentage (e.g. "94% Match").



Pillar 3: Sprint Workspace Kanban

Interactive 4-column sprint task board (To Do, In Progress, Review, Done) with drag-and-drop task movement, priority tags, and assignees.

System Architecture & Execution Flow

How data flows seamlessly from student skill profiles to completed project deliverables:

1

Profile Creation

Students register technical skills, proficiencies, GitHub/portfolio links, and team desires.

2

Match Calculation

Synergy algorithm computes complementary fit between student profile and open project roles.

3

Recruit & Accept

Leader reviews applicant pitch in the Hub and clicks 'Accept', adding the student to the active roster.

4

Kanban Sprint

Tasks are assigned to team members and moved through sprint columns until final submission.

Entity-Relationship (ER) Schema

Relational database structure governing users, projects, roles, applications, and sprint tasks:

USER / STUDENT (1)

```
PK user_id: VARCHAR
name: VARCHAR
academic_role: VARCHAR
university: VARCHAR
skills: JSON ARRAY
reputation_score: INT
```

PROJECT (M)

```
PK project_id: VARCHAR
FK leader_id: VARCHAR
title: VARCHAR
category: VARCHAR
deadline: DATE
sprint_progress: INT
```

OPEN_ROLE (M)

```
PK role_id: VARCHAR
FK project_id: VARCHAR
role_name: VARCHAR
skill_needed: VARCHAR
is_filled: BOOLEAN
```

APPLICATION (M)

```
PK application_id: VARCHAR
FK project_id: VARCHAR
FK applicant_id: VARCHAR
applied_role: VARCHAR
pitch_message: TEXT
status: ENUM
```

SPRINT_TASK (M)

```
PK task_id: VARCHAR
FK project_id: VARCHAR
title: VARCHAR
status: ENUM
priority: ENUM
assignee_name: VARCHAR
```

RELATIONSHIPS

- USER **leads (1:N)** PROJECT
- PROJECT **requires (1:N)** OPEN_ROLE
- USER **submits (1:N)** APPLICATION
- PROJECT **tracks (1:N)** SPRINT_TASK
- USER **assigned (M:N)** SPRINT_TASK

Complementary Skill Matchmaking Algorithm

How SkillBridge prevents team skill imbalance and encourages multidisciplinary projects:

```
// Synergy Scoring Calculation
Score = 0; TotalChecks = 0;

For each role in project.openRoles:

  TotalChecks += 2.0;
  If student.hasSkill(role.skillNeeded):
    Score += 2.0; // Direct Fit
  Else If isComplementary(student, role):
    Score += 1.5; // High Synergy Pair
  Else:
    Score += 0.8; // Academic Baseline

MatchRate = min(98%, round(Score/TotalChecks * 100));
```

Direct Role Alignment (+2.0)

When an applicant's core expertise matches the exact unfilled technical role advertised by the leader.

Complementary Synergy (+1.5)

Rewards cross-discipline pairings (e.g. UI/UX Designer paired with a Backend system, or ML specialist with Data engineering).

Zero Skill Saturation

Discourages duplicate skill applications to ensure balanced teams with end-to-end capabilities.

Core System Modules

Four fully implemented and interconnected academic workflows:

1. Explore & Project Discovery Feed

Real-time search by keyword, domain category filters (AI/ML, Mobile, Web3), dynamic skill match percentage badges, and rapid one-click application modal.

2. Team Leader Portal

Review incoming student applications, read custom pitch messages, inspect portfolio links, and accept or decline candidates with immediate roster sync.

3. Project Workspace Kanban Board

Interactive 4-column sprint board (To Do, In Progress, Review, Done) with drag-and-drop task movement, priority pills, assignees, and due dates.

4. Student Profile & Persona Switcher

Technical skill proficiency bars, complementary role desires, and a live persona switcher allowing evaluators to test as different student roles.

Technical Stack & System Requirements

Pure vanilla web standards ensuring zero dependency issues, high performance, and rapid deployment:

Software Technologies

- **Frontend Structure:** Semantic HTML5 & accessible modals
- **Styling:** Vanilla CSS3 (Custom design system, glassmorphism)
- **Application Logic:** Modern JavaScript (ES6+), Event-driven SPA
- **Data Persistence:** Web Storage API (localStorage)
- **Interactivity:** HTML5 Drag-and-Drop Kanban API

Hardware & Deployment

- **Processor:** Intel Core i3 (4th Gen) / AMD Ryzen 3 or higher
- **RAM:** 4 GB minimum (8 GB recommended)
- **Browser Support:** Google Chrome, Edge, Firefox, Brave
- **Local Server:** Python HTTP Server (`python -m http.server 3000`)
- **Zero Build Step:** Instant startup without npm dependency bloat

Roadmap: Upgrade to Major Project

Modular design enables effortless scaling from minor project to enterprise campus deployment:

1. Full-Stack Cloud Database

Migrate client LocalStorage to Node.js/Express backend with PostgreSQL / MongoDB and Prisma ORM for multi-user synchronization.

2. Real-Time Workspace Chat

Integrate WebSockets (Socket.io) inside the project workspace for live team discussion, file uploads, and milestone notifications.

3. University Single Sign-On (SSO)

Authenticate student registration through Institutional Google Workspace / LDAP using College Roll Numbers.

4. AI Synopsis & Plagiarism Checker

Automated verification against past academic batches to ensure project uniqueness and assist in drafting synopses.

Conclusion & Project Defense

SkillBridge eliminates the barriers student innovators face by bridging project discovery, complementary skill matchmaking, and team task tracking in one unified platform.

Key Academic Achievements:

- Solved team skill redundancy using the Complementary Fit algorithm.
- Integrated full lifecycle support from idea pitch to Sprint Kanban delivery.
- Validated complete state persistence and zero-dependency cross-browser execution.

Thank You! We Welcome Questions & Feedback.